Year 11 Mathematics

Number 8

1.4 Ratio

Ratio

STARTER · 5 MIN

Date: _____ **Title:** Ratio

LEARNING OBJECTIVES

- Simplify ratios
- Share an amount in a given ratio
- Solve ratio problems

STARTER *Answer in your book. No calculators.*

1. Simplify $\frac{12}{18}$
2. Work out $35 \div 7 \times 3$
3. Find the HCF of 24 and 40
4. Convert 2 m into cm

Starter — Answers

STARTER

Tick or correct in green pen

1. $\frac{12}{18} = \frac{\mathbf{2}}{\mathbf{3}}$

2. $5 \times 3 = \mathbf{15}$

3. HCF = **8**

4. $2 \times 100 = \mathbf{200\ cm}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can simplify ratios	☐
SILVER	I can share an amount in a given ratio	☐
GOLD	I can solve ratio problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Ratio

INTRODUCTION · 3 MIN

A ratio compares the sizes of two or more quantities.

Simplify

Divide all parts by the HCF.

12 : 18 = 2 : 3

Same units

Change to the same units first.

50 cm : 2 m = 50 : 200 = 1 : 4

Sharing

Add the parts, find one part, then multiply.

£40 in 3 : 5 → 1 part = £5

One part known

Use it to find the others.

3 parts = 12, so 1 part = 4

Examples 1 and 2 — Simplifying Ratios

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Simplify 15 : 25

Example 2

Simplify 40 cm : 1.2 m

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

HCF = 5

15 : 25 = 3 : 5

Tip: divide both parts by the HCF.

Example 2

1.2 m = 120 cm

40 : 120 = 1 : 3

Tip: use the same units.

Task 1

PRACTICE · 9 MIN

BRONZE *Simplify each ratio*

1. 8 : 12
2. 20 : 45
3. 6 : 9 : 15
4. 300 g : 2 kg
5. 45 min : 2 hours

★ Write 4 : 10 in the form 1 : n

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $8 : 12 = 2 : 3$

2. $20 : 45 = 4 : 9$

3. $6 : 9 : 15 = 2 : 3 : 5$

4. $300 : 2000 = 3 : 20$

5. $45 : 120 = 3 : 8$

★ $4 : 10 = 1 : 2.5$

Examples 3 and 4 — Sharing in a Ratio

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint 1 part = total ÷ number of parts

Example 3

Share £60 in the ratio 2 : 3.

Example 4

Share 84 sweets in the ratio 1 : 2 : 4.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

2 + 3 = 5 parts

1 part = 60 ÷ 5

= 12

£24 and £36

Tip: check: 24 + 36 = 60.

Example 4

1 + 2 + 4 = 7 parts

1 part = 84 ÷ 7

= 12

12, 24 and 48

Tip: multiply each part by 12.

Task 2

PRACTICE · 9 MIN

SILVER

Share each amount in the ratio given

Hint 1 part = total ÷ number of parts

1. £50 in the ratio 3 : 2
2. £72 in the ratio 5 : 1
3. 90 kg in the ratio 4 : 5
4. £120 in the ratio 1 : 2 : 3
5. 200 ml in the ratio 3 : 7

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 1 part = £10, **£30 and £20**
2. 1 part = £12, **£60 and £12**
3. 1 part = 10 kg, **40 kg and 50 kg**
4. 1 part = £20, **£20**, **£40 and £60**
5. 1 part = 20 ml, **60 ml and 140 ml**

Examples 5 and 6 — Ratio Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint 1 part = total ÷ number of parts

Example 5

(2 marks)

Boys and girls are in the ratio 4 : 5.

There are 20 boys. How many girls are there?

Example 6

(3 marks)

Ali and Sara share money in the ratio 3 : 7.

Sara gets £28 more than Ali. How much does Ali get?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

4 parts = 20

1 part = 5

Girls = 5 × 5

= 25

Tip: find one part first.

Example 6

7 − 3 = 4 parts

= £28

1 part = £7

Ali = 3 × 7

= £21

Tip: the difference is 4 parts.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint 1 part = total ÷ number of parts

1. Red and blue counters are in the ratio 2 : 7. There are 14 red. How many blue? (2 marks)

2. A recipe for 4 people uses 300 g flour. How much for 10 people? (2 marks)

3. Cement and sand are mixed 1 : 4. How much sand is in 35 kg of mix? (2 marks)

4. Tom and Lina share sweets 5 : 3. Tom gets 12 more than Lina. How many sweets in total? (3 marks)

5. x : y = 3 : 4 and y : z = 2 : 5. Find x : y : z. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. 1 part = 7, Blue = **49**

2. $300 \div 4 \times 10 =$ **750 g**

3. 1 part = 7 kg, Sand = **28 kg**

4. 2 parts = 12, 1 part = 6, Total = **48**

5. $y : z = 4 : 10$, **$x : y : z = 3 : 4 : 10$**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify 18 : 24

SILVER Share £45 in the ratio 4 : 5

GOLD Paint is mixed 2 : 3. 10 litres of the first colour are used. How much of the second?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **18 : 24 = 3 : 4**

SILVER 1 part = £5, **£20 and £25**

GOLD 1 part = 5, **15 litres**

Exam Practice — Simplifying Ratios

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

Write the ratio 24 : 36 in its simplest form.

.....

Question 2 (2 marks)

Write 750 g : 2 kg in its simplest form.

.....

Exam Practice — Sharing in a Ratio

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint 1 part = total ÷ number of parts

Question 3 (2 marks)

Share £84 in the ratio 3 : 4

.....

Question 4 (3 marks)

Ann, Ben and Cal share £180 in the ratio 2 : 3 : 5

How much does Cal get?

..... £

Exam Practice — Ratio Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint 1 part = total ÷ number of parts

Question 5

(3 marks)

Red and blue beads are in the ratio 5 : 2

There are 45 more red beads than blue beads.

How many beads are there altogether?

.....

Question 6

(3 marks)

a : b = 2 : 3 and b : c = 4 : 5

Find a : c

Give your answer in its simplest form.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

HCF of 24 and 36 = 12 — M1

= 2 : 3 — A1

Tip: divide both parts by the HCF.

Question 2

2 kg = 2000 g — M1

750 : 2000 = 3 : 8 — A1

Tip: change to the same units first.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

7 parts: 84 ÷ 7 = 12 — M1

£36 and £48 — A1

Tip: check: 36 + 48 = 84.

Question 4

2 + 3 + 5 = 10 parts — M1

1 part = 180 ÷ 10 = 18 — M1

Cal: 5 × 18 = £90 — A1

Tip: add the parts first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

5 − 2 = 3 parts = 45	M1
1 part = 15	M1
7 × 15 = 105	A1

Tip: the difference is 3 parts.

Question 6

a : b = 8 : 12, b : c = 12 : 15	M1
a : b : c = 8 : 12 : 15	M1
a : c = 8 : 15	A1

Tip: make the b parts the same.